

Impact of Granulation Solvent on Release Performance of Dual Matrix Extended-Release Tablets of Highly Water-Soluble Drugs Formulated using Compritol® 888 ATO and Methocel® as Release Retarding Agents

Ketkee Deshmukh, Swarnadeep Banerjee, Agnivesh Shrivastava, Sunil Bambarkar
Gattefosse India Pvt. Ltd., Mumbai, India

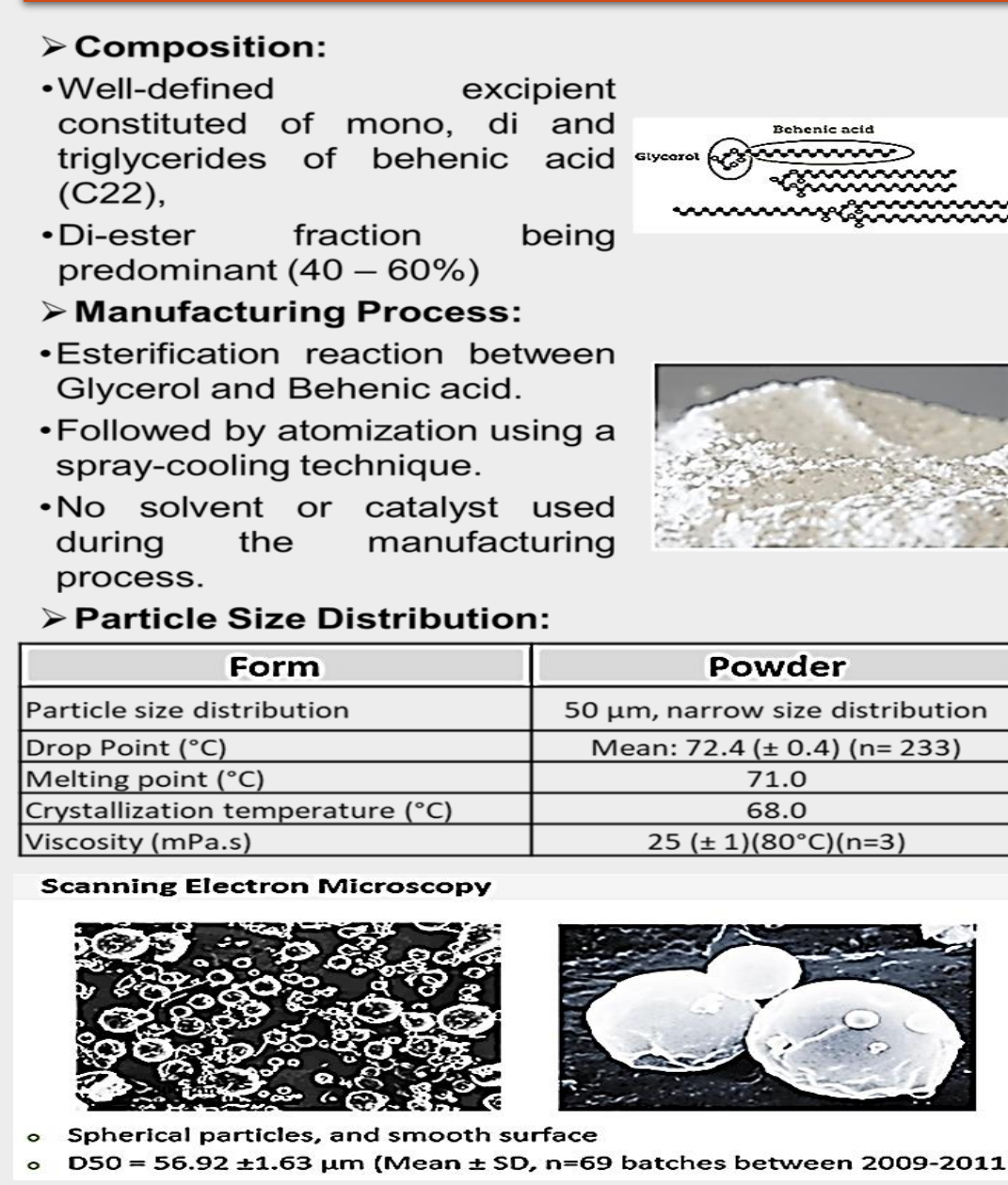
CONTACT INFORMATION: kdeshmukh@gattefosse.in



PURPOSE

- Combination of Compritol® 888 ATO and HPMC polymers as Extended Release (ER) dual matrix forming agents aids in better controlled release of highly water-soluble drugs.
- Type of granulation solvent used during wet granulation process affects the degree of hydration of HPMC polymers impacting the granule, tablet properties and release of drug from the matrix.
- Purpose of this study is to evaluate the impact of different granulation solvents on the performance of highly water-soluble high dose Metformin Hydrochloride 1000 mg dual matrix ER tablets USP and of freely water-soluble low dose Metoprolol Succinate 95 mg dual matrix ER tablets USP formulated using combination of Compritol® 888 ATO and Methocel® K100M as ER dual matrix forming agents by wet granulation process.

Fig. 1: Compritol® 888 ATO at a glance



OBJECTIVE

To study the impact of three granulation solvents: deionized water (DW), isopropyl alcohol (IPA) and DW: IPA (1:1) mixture on the performance of Metformin Hydrochloride 1000 mg dual matrix ER tablets USP and Metoprolol Succinate 95 mg dual matrix ER tablets USP formulated using combination of Compritol® 888 ATO and Methocel® K100M as ER dual matrix forming agents by wet granulation process.

METHODS

Table 1: Formula for Metformin Hydrochloride 1000mg ER tablets USP/NF and Metoprolol Succinate 95mg ER tablets USP used in the study

Formula for Metformin Hydrochloride 1000 mg ER dual matrix tablets USP				Formula for Metoprolol Succinate 95 mg ER dual matrix tablets USP			
Ingredients	% w/w	Mg/tab	Role of ingredient in formula	Ingredients	% w/w	Mg/tab	Role of ingredient in formula
Intragranular				Intragranular			
Metformin Hydrochloride	71.43	1000.000	API	Metoprolol Succinate	15.83	95.000	API
Compritol® 888 ATO	2.86	40.000	ER Matrixing Agent	Compritol® 888 ATO	19	114.000	ER Matrixing Agent
Methocel® K100 M	17.00	238.000	ER Matrixing Agent	Methocel® K4M	15.83	95.000	ER Matrixing Agent
Dicalcium Phosphate	3.00	42.000	Diluent	Methocel® K100M CR	22.17	133.000	ER Matrixing Agent
PVP K90	3.00	42.000	Binder	Avicel® PH 101	2.5	15.000	Binder
Extragranular				Extragranular			
Aerosil® 200 Pharma	0.71	10.000	Gilant	Prosolv® SMCC 90 HD	6.25	37.500	Extragranular Diluent
Compritol® 888 ATO	1.00	14.000	ER Matrixing Agent	Emcompress® Premium	2.67	16.000	Extragranular Diluent
Magnesium Stearate	1.00	14.000	Lubricant	Sodium Stearyl Fumarate	1.08	6.500	Lubricant
Total	100.00	1400.000		Total	100.00	600.000	

Fig. 2: Schematic Representation of Manufacturing Process used in the study

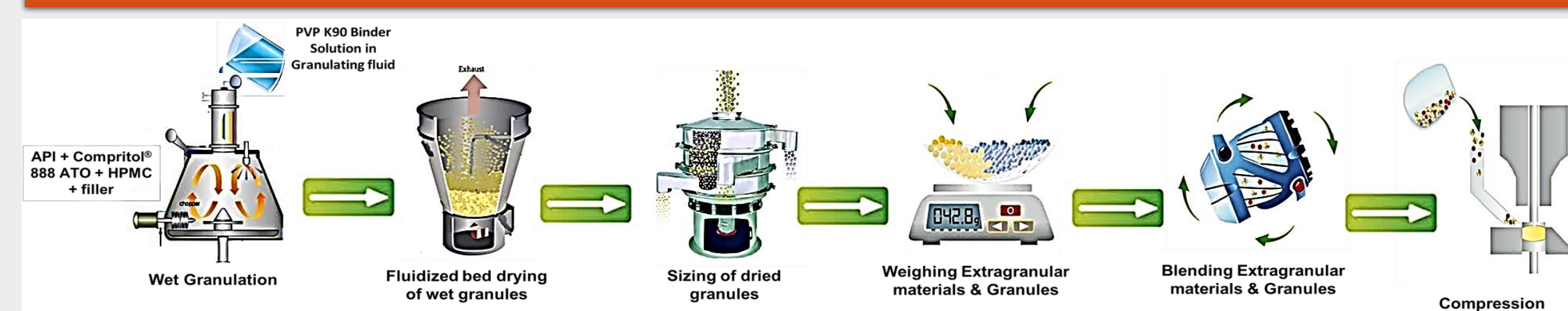


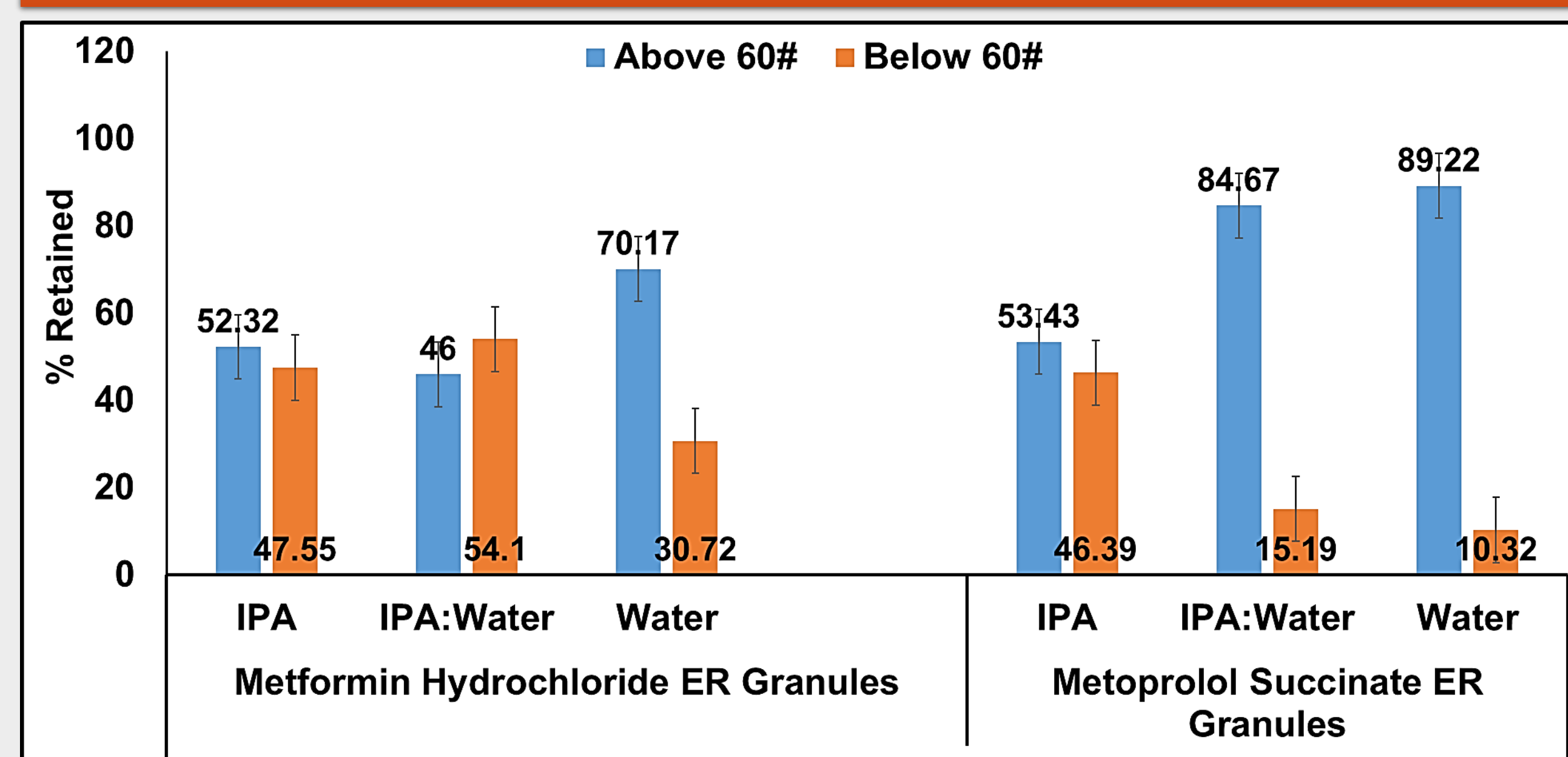
Fig. 3: Evaluating impact of Granulation solvents on Granule & Tablet properties of Metformin Hydrochloride 1000mg ER tablets USP and Metoprolol Succinate 95mg ER tablets USP

Granule Property	- Particle Size Distribution - Flow Properties - Granule Density
Tablet Property	- Tablet Hardness - Tensile Strength - Swelling Index
Drug Release Kinetics	- Dissolution Test 4 specified in USP41/NF36 monograph for Metformin Hydrochloride 1000mg ER Tablets USP/NF - Dissolution Test 1 specified in USP41/NF36 monograph for Metoprolol Succinate 95mg ER tablets USP/NF.

RESULTS

Evaluation of Granule Properties:

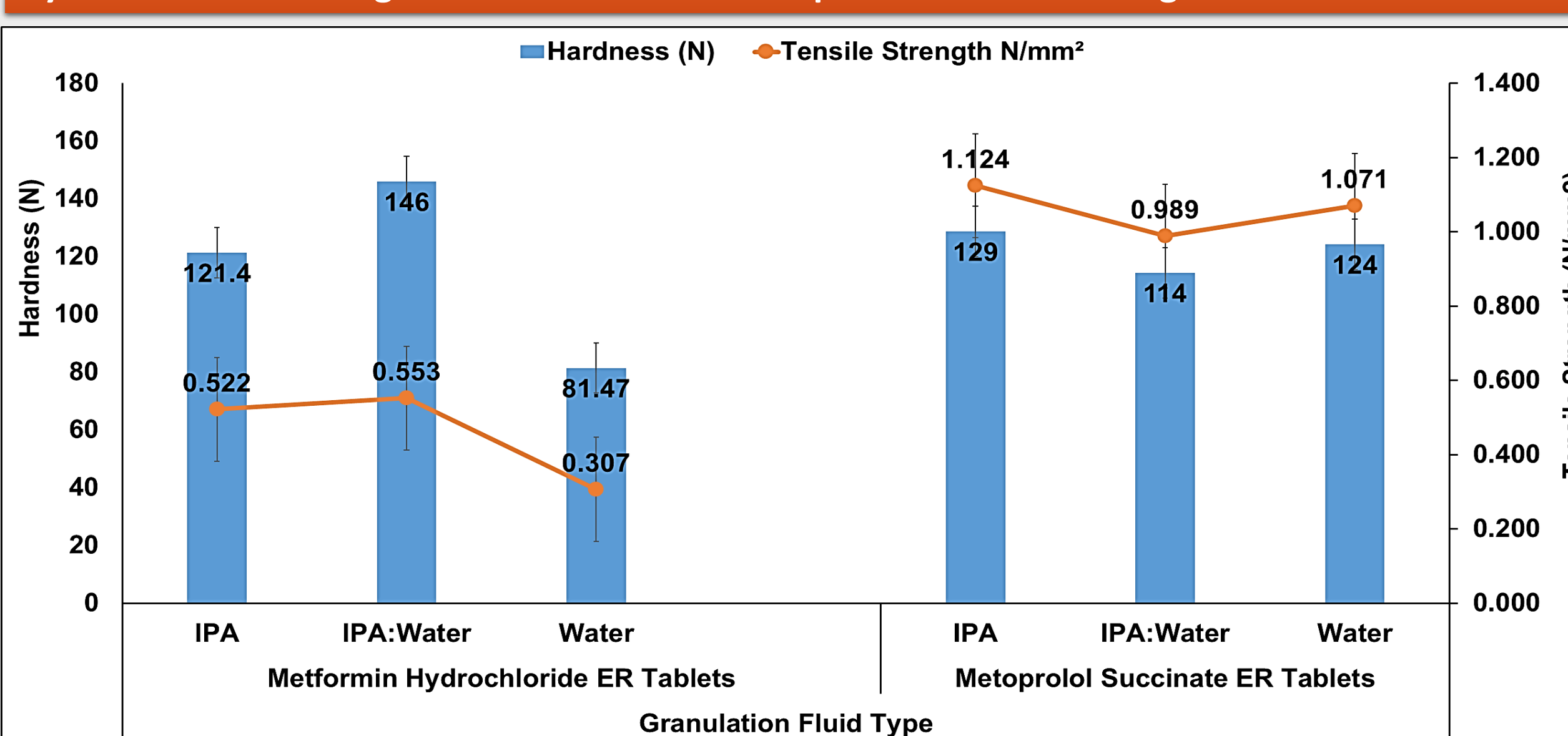
Fig. 4. Particle Size Distribution of granules of Metformin Hydrochloride 1000mg ER tablets USP and Metoprolol Succinate 95mg ER tablets USP



Observation: Better control on particle size distribution was observed with granules formulated using DW as granulating solvent in comparison to IPA & DW: IPA (1:1) mixture owing to better hydration of Methocel®K100M in both formulations.

Evaluation of Tablet Properties:

Fig. 5. Comparison of max. hardness & Tensile Strength achieved for tablets of Metformin Hydrochloride 1000mg ER tablets USP and Metoprolol Succinate 95mg ER tablets USP



Observation:

- Metformin Hydrochloride 1000 mg ER tablets formulated using IPA:DW (1:1) mixture showed higher hardness and Tensile strength in comparison to plain IPA & DW.
- Metoprolol Succinate 95 mg ER tablets formulated using plain IPA & plain DW showed higher hardness & Tensile Strength in comparison to those formulated using IPA:Water (1:1) mixture.
- As per Swelling Index study, Metformin Hydrochloride ER Tablets formulated using IPA:DW (1:1) mixture showed slightly higher swelling in comparison to the tablets manufactured using plain IPA & plain DW.
- No difference in the swelling behaviour was observed for Metoprolol Succinate ER Tablets formulated using the three granulating solvents under study.
- Drug release profile of Metformin Hydrochloride ER Tablets formulated using IPA:DW (1:1) mixture complied with the Dissolution Test 4 specified in USP41/NF36 monograph for Metformin Hydrochloride 1000mg ER Tablets USP. The other two variants failed to comply.
- All the Metoprolol Succinate 95 mg ER tablets formulated using the three granulating fluid under study complied with Dissolution Test 1 specified in USP41/NF36 monograph for Metoprolol Succinate 95mg ER tablets USP.

Table 2. Flow Properties & density of granules of Metformin Hydrochloride 1000 mg ER tablets USP

Granulating fluid	Metformin Hydrochloride 1000 mg ER tablets USP		Flow Properties	
	Granule Density			
	Bulk Density	Tap Density	Hausner's ratio	Carr's index %
PVP K90 + (IPA : Water)	0.44 g/mL	0.57 g/mL	1.28	Passable
PVP K90 + Water	0.51 g/mL	0.64 g/mL	1.25	Fair
PVP K90 + IPA	0.44 g/mL	0.55 g/mL	1.25	Fair

Observation: DW & IPA as granulating solvent overall proved to be effective in improving flow properties of Metformin Hydrochloride ER granules

Table 3. Flow Properties & density of granules of Metoprolol Succinate 95 mg ER tablets USP

Granulating fluid	Metoprolol Succinate 95 mg ER tablets USP		Flow Properties	
	Granule Density			
	Bulk Density	Tap Density	Hausner's ratio	Carr's index %
PVP K90 + (IPA : Water)	0.41 g/mL	0.57 g/mL	1.37	Poor
PVP K90 + Water	0.38 g/mL	0.5 g/mL	1.3	Passable
PVP K90 + IPA	0.36 g/mL	0.46 g/mL	1.27	Passable

Observation: DW & IPA as granulating solvents proved to be effective in improving flow properties of Metoprolol Succinate ER granules

Fig. 6. Comparison of Swelling Index of Metformin Hydrochloride 1000 mg ER tablets USP and Metoprolol Succinate 95 mg ER tablets USP in respective dissolution mediums

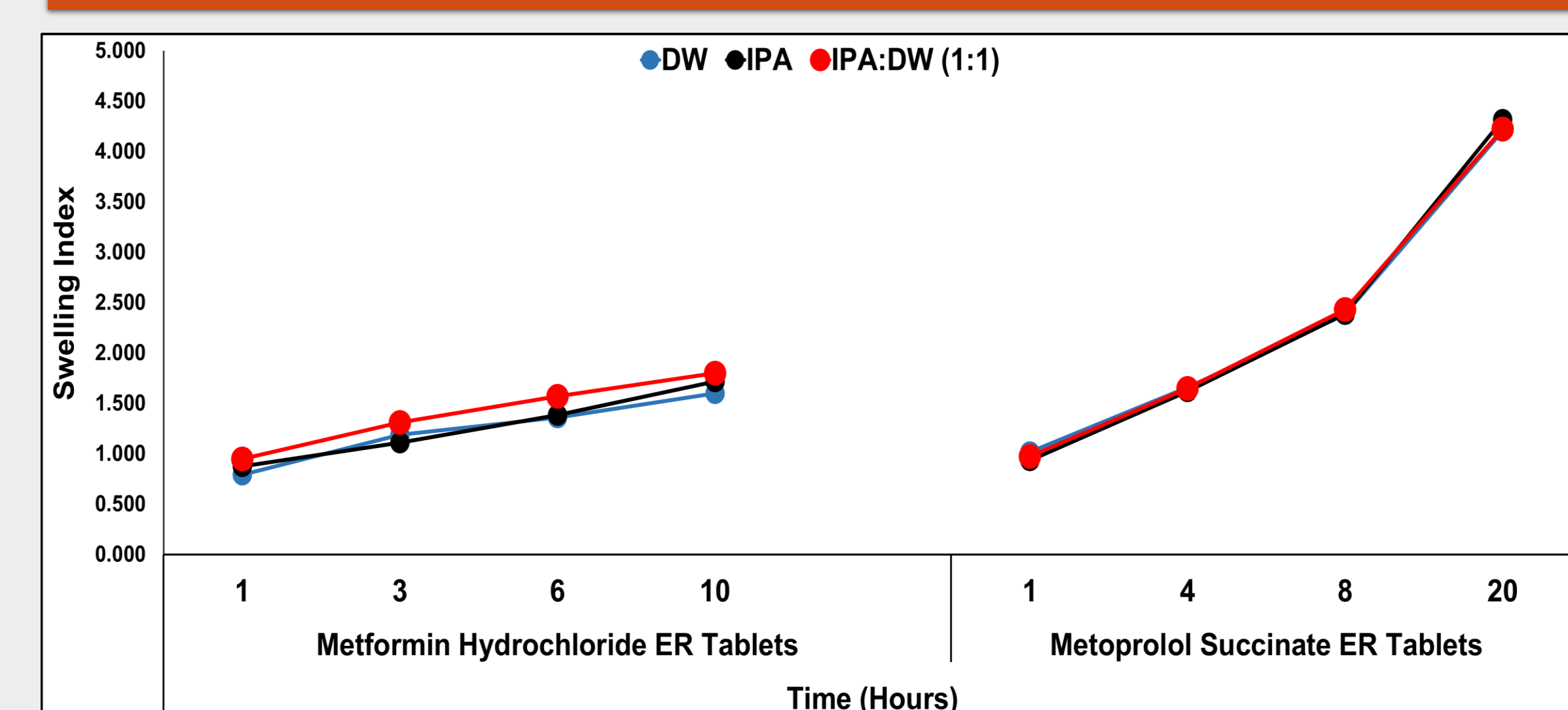
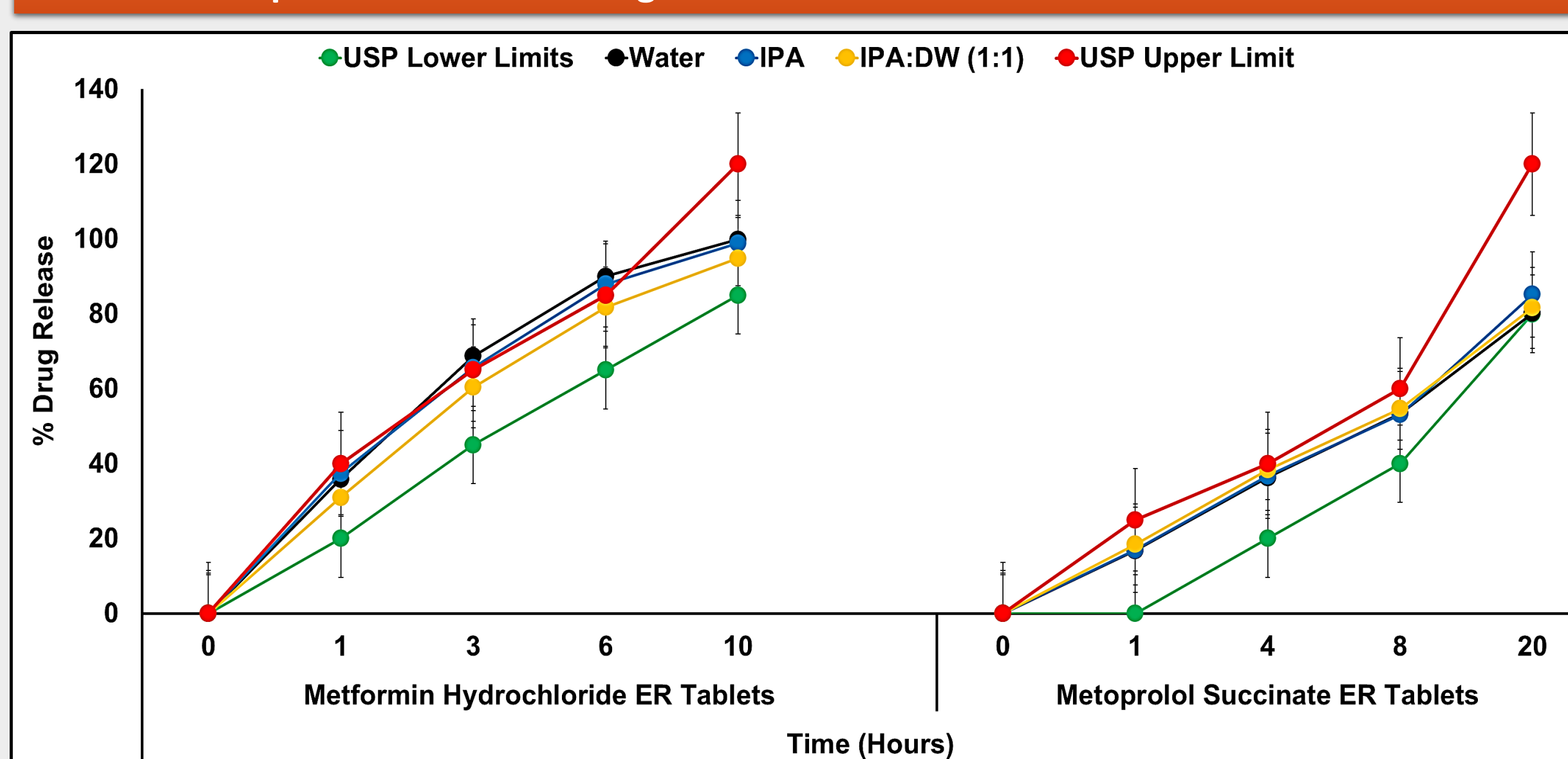


Fig. 7. Comparison of Drug Release Kinetics of Metformin Hydrochloride 1000mg ER tablets USP and Metoprolol Succinate 95 mg ER tablets USP



CONCLUSION

- Owing to high dose of Metformin Hydrochloride 1000 mg ER dual matrix tablets USP, the API occupied more than 70% of the matrix space while the concentration of both release controlling agents was restricted to about 20%.
- Slightly higher swelling of the ER matrix afforded by the IPA:DW (1:1) mixture resulted in compliance of the Metformin Hydrochloride ER Tablets with the Dissolution Test 4 specified in USP41/NF36 monograph for Metformin Hydrochloride 1000mg ER Tablets USP.
- Metoprolol Succinate being freely water soluble required considerable amount of release retarding agents to achieve desired release profile wherein the API occupied 15% of the tablet matrix while the dual release retarding agents constituted to 57% of total tablet weight.
- Owing to this high polymer content and low API concentration in the Metoprolol Succinate ER matrix, no impact of matrix hydration by the granulation solvent was observed on the swelling behaviour and drug release kinetics.
- Thus, it can be concluded that the choice of granulation solvent impacts release performance of dual matrix extended-release tablets formulated using Compritol® 888 ATO and Methocel® as release retarding agents and is governed by: percentage of API occupying the ER matrix & concentration of the release retarding polymer forming the matrix.

FUNDING/GRANT/ENCORE/REFERENCE OR OTHER USE

Promote Your Poster!

AAPS allows you to contact attendees before and during the meeting through its app. Invite a key contact to your presentation! AAPS also posts posters online after the event, where your work can generate additional contacts for you.

